Software Requirements Specification

StuDEX: Department-wise Achievement Tracker

Version 1.0

Prepared by: Pawas Pandey

ABES Engineering College

April 1 1, 2025

Table of Contents

---

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document outlines the requirements for "StuDEX," a department-wise achievement tracker web application. It provides a detailed description of the system's features, interfaces, and behaviors, serving as a reference for development and evaluation.

1.2 Document Conventions

This document follows IEEE SRS format guidelines. Requirements are categorized as:

- FR: Functional Requirements
- NFR: Non-Functional Requirements

Priority levels are indicated as:

- High: Essential feature
- **Medium:** Important but not critical

- Low: Desirable but can be implemented later

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Project developers and technical team
- Project supervisors and evaluators
- Department administrators who will use the system
- Testers responsible for system validation

Developers should focus on sections 3 and 4, while stakeholders may focus on sections 1, 2, and 5.

1.4 Project Scope

StuDEX is a web-based application designed to track and showcase achievements of students across different departments within the college. The system will facilitate achievement submission, verification, department-wise analytics, and public showcasing of accomplishments.

The system aims to:

- Streamline the process of recording student achievements
- Provide statistical insights on department performances
- Create a digital repository of student accomplishments
- Foster healthy competition among departments
- Showcase department and student achievements to the public

1.5 References

- IEEE Standard 830-1998, IEEE Recommended Practice for Software Requirements Specifications
- React.js Documentation: https://reactjs.org/docs/getting-started.html
- Next.js Documentation: https://nextjs.org/docs
- PostgreSQL Documentation: https://www.postgresql.org/docs/
- MongoDB Documentation: https://docs.mongodb.com/

2. Overall Description

2.1 Product Perspective

StuDEX is a standalone web application that will integrate with the existing college information system. It will serve as a centralized platform for tracking student achievements across all departments. The system will be accessible to students, faculty, administrators, and the public with appropriate access controls.

2.2 Product Functions

The major functions of StuDEX include:

- User registration and authentication
- Student profile management
- Achievement submission and management
- Achievement verification workflow
- Department-wise analytics and reporting
- Search and filter capabilities
- Administrative functions for system management
- Public showcase of achievements
- Notification system for updates and alerts

2.3 User Classes and Characteristics

1. **Students:**
   - Primary users who will submit and track their achievements
   - Moderate technical proficiency expected
   - Frequent access to the system
2. **Faculty Members:**
   - Verify student achievements within their departments
   - Generate reports and view department analytics
   - Moderate technical proficiency expected
   - Regular access to the system
3. **Department Administrators:**
   - Manage department data and users
   - Monitor department performance metrics
   - Higher technical proficiency expected
   - Regular access to the system
4. **System Administrators:**
   - Manage the entire system and user access
   - Configure system parameters

- High technical proficiency required
- Occasional access for maintenance

5. Public Visitors:
    - View publicly available achievement showcases
    - No technical proficiency required
    - Occasional access to the system

2.4 Operating Environment

- Web Browsers: Chrome (v90+), Firefox (v88+), Safari (v14+), Edge (v90+)
- Devices: Desktop computers, laptops, tablets, and mobile phones
- Operating Systems: Windows, macOS, Linux, Android, iOS
- Concurrent Users: Up to 500 simultaneous users
- Network: Standard internet connection (minimum 1 Mbps)

2.5 Design and Implementation Constraints

- **Technology Stack:**
    - Frontend: React/Next.js with CSS
    - Backend: MERN (MongoDB, Express.js, React.js, Node.js)
    - Database: PostgreSQL
- Development Timeline: To be completed within the academic semester
- Hosting: College server infrastructure
- Security Compliance: Must adhere to educational data privacy standards
- Accessibility: Must meet WCAG 2.1 Level AA requirements

2.6 User Documentation

The following user documentation will be delivered with the system:

- Online help system within the application
- User manuals for each user class
- Installation and administration guide
- Video tutorials for common tasks
- Frequently Asked Questions (FAQ) section

2.7 Assumptions and Dependencies

- Users have access to devices with internet connectivity
- The college provides server infrastructure for hosting
- API access to department information system (if integration is required)
- Students and faculty have basic computer literacy
- Email services are available for notifications

3. System Features

3.1 User Authentication System

3.J.J Description

The system shall provide secure authentication mechanisms for different user roles.

3.1.2 Functional Requirements

FR-1.1 (High): The system shall allow users to register with email, name, student ID, and department.

FR-J.2 (High): The system shall provide secure login functionality with email/username and password.

FR-1.3 (Medium): The system shall implement role-based access control (student, faculty, admin).

FR-1.4 (Medium): The system shall provide password recovery mechanism through email.

FR-J.5 (Low): The system shall support multi-factor authentication for administrative accounts.

FR-J.6 (Medium): The system shall maintain user session management with appropriate timeouts.

3.2 User Profile Management

3.2.J Description

Users shall be able to create and manage their profiles with personal and academic information.

3.2.2 Functional Requirements

FR-2.1 (High): The system shall allow users to create profiles with basic information (name, department, academic year).

FR-2.2 (Medium): The system shall allow users to upload profile pictures.

FR-2.3 (Medium): The system shall enable users to update their profile information.

FR-2.4 (Medium): The system shall display a user's achievement summary on their profile.

FR-2.5 (Low): The system shall allow users to set privacy preferences for their achievements.

FR-2.6 (Low): The system shall enable users to link social media accounts.

3.3 Achievement Management

3.3.J Description

The core functionality allowing students to submit, track, and manage their achievements.

3.3.2 Functional Requirements

FR-3.J (High): The system shall provide a structured form for achievement submission.

FR-3.2 (High): The system shall allow categorization of achievements (academic, sports, cultural, research).

FR-3.3 (High): The system shall enable file uploads for achievement evidence/certificates.

FR-3.4 (High): The system shall implement approval workflow for achievement verification.

FR-3.5 (Medium): The system shall allow students to edit or delete pending achievements.

FR-3.6 (Medium): The system shall track the status of achievement verifications.

FR-3.7 (Medium): The system shall notify users of approval status changes.

FR-3.8 (Low): The system shall support achievement tagging for better categorization.

3.4 Dashboard and Analytics

3.4.J Description

Interactive dashboards displaying achievement metrics and analytics for individuals and departments.

3.4.2 Functional Requirements

FR-4.J (High): The system shall provide individual achievement dashboards for students.

FR-4.2 (High): The system shall display department-wise achievement statistics.

FR-4.3 (Medium): The system shall generate visual representations of achievement data (charts, graphs).

FR-4.4 (Medium): The system shall implement leaderboards for departments and individuals.

FR-4.5 (Medium): The system shall allow comparison of achievement metrics between departments.

FR-4.6 (Low): The system shall provide trend analysis of achievements over time.

FR-4.7 (Low): The system shall allow customization of dashboard views.

3.5 Notification System

3.5.1 Description

System-generated alerts and notifications for relevant events and updates.

3.5.2 Functional Requirements

FR-5.1 (High): The system shall send notifications for achievement status updates.

FR-5.2 (Medium): The system shall provide in-app notification center.

FR-5.3 (Medium): The system shall send email notifications for critical updates.

FR-5.4 (Low): The system shall allow users to configure notification preferences.

FR-5.5 (Low): The system shall provide reminders for pending verifications to faculty.

3.6 Search and Filter Functionality

3.6.1 Description

Features allowing users to search and filter achievements based on various criteria.

3.6.2 Functional Requirements

FR-6.1 (High): The system shall provide basic search functionality for achievements.

FR-6.2 (Medium): The system shall implement advanced filters (department, category, date range).

FR-6.3 (Medium): The system shall support sorting of search results.

FR-6.4 (Low): The system shall provide saved search preferences.

FR-6.5 (Low): The system shall implement autocomplete for search queries.

3.7 Administration Panel

3.7.1 Description

Administrative interface for system management and configuration.

3.7.2 Functional Requirements

FR-7.1 (High): The system shall provide user management capabilities (create, edit, disable).

FR-7.2 (High): The system shall allow management of achievement categories.

FR-7.3 (Medium): The system shall provide system configuration options.

FR-7.4 (Medium): The system shall allow bulk operations on achievements and users.

FR-7.5 (Medium): The system shall provide audit logs of administrative actions.

FR-7.6 (Low): The system shall allow customization of system-wide settings.

3.8 Department Management

3.8.1 Description

Features for managing department information and structure within the system.

3.8.2 Functional Requirements

FR-8.J (High): The system shall allow creation and management of department profiles.

FR-8.2 (Medium): The system shall associate faculty members with departments.

FR-8.3 (Medium): The system shall enable department-specific announcements.

FR-8.4 (Medium): The system shall provide department-specific achievement settings.

FR-8.5 (Low): The system shall support department hierarchy if applicable.

4. External Interface Requirements

4.1 User Interfaces

UI-1: The application shall utilize a responsive design for compatibility with different devices.

UI-2: The primary user interface shall include:

- Home/landing page with overview statistics
- User authentication screens (login/registration)
- User profile interface
- Achievement submission and management interface
- Dashboard and analytics interface
- Search and filter interface
- Administration panel (for authorized users)
- Department management interface (for authorized users)

UI-3: The UI shall comply with WCAG 2.1 Level AA accessibility guidelines.

UI-4: The system shall provide consistent navigation across all pages.

UI-5: Form validations shall provide clear feedback to users.

4.2 Hardware Interfaces

HI-1: The system shall be compatible with standard input devices (keyboard, mouse, touchscreen).

HI-2: The system shall support file uploads from various storage devices.

HI-3: The system shall be optimized for both desktop and mobile device usage.

4.3 Software Interfaces

SI-J: The system shall interface with PostgreSQL database system for data storage.

SI-2: The system shall utilize React/Next.js for frontend rendering.

SI-3: The system shall implement RESTfuI API architecture for client-server communication.

SI-4: The system shall potentially interface with college information system API (if available).

SI-5: The system shall support integration with email service providers for notifications.

4.4 Communications Interfaces

CI-1: The system shall use HTTPS protocol for secure data transmission.

CI-2: The system shall handle WebSocket connections for real-time notifications.

CI-3: The system shall implement appropriate error handling for network issues.

CI-4: The system shall support standard internet connectivity requirements.

5. Non-functional **Requirements**

5.1 Performance Requirements

NFR-J.J: The system shall load pages within 3 seconds under normal network conditions.

NFR-J.2: The system shall support at least 500 concurrent users without performance degradation.

NFR-J.3: The system shall handle at least 100 achievement submissions per minute.

NFR-J.4: Database queries shall complete within 2 seconds.

NFR-J.5: File uploads shall support files up to 10MB in size.

5.2 Safety Requirements

NFR-2.J: The system shall maintain data integrity during concurrent operations.

NFR-2.2: The system shall implement validation checks to prevent data corruption.

NFR-2.3: The system shall perform regular data backups to prevent data loss.

5.3 Security Requirements

NFR-3.1: The system shall encrypt all passwords using strong hashing algorithms.

NFR-3.2: The system shall implement HTTPS for all data transmissions.

NFR-3.3: The system shall enforce session timeouts after 30 minutes of inactivity.

NFR-3.4: The system shall validate all user inputs to prevent injection attacks.

NFR-3.5: The system shall implement access controls based on user roles.

NFR-3.6: The system shall maintain audit logs for security-sensitive operations.

5.4 Software Quality Attributes

NFR-4.1: Usability - The system shall be intuitive enough that a new user can submit an achievement within 5 minutes of registration.

NFR-4.2: Reliability - The system shall have an uptime of at least 99.5%.

NFR-4.3: Maintainability - The system shall follow modular design principles for ease of maintenance.

NFR-4.4: Portability - The system shall work consistently across specified web browsers and devices.

NFR-4.5: Scalability - The system shall be designed to accommodate growth in users and data volume.

5.5 Business Rules

NFR-5.J: Only faculty members can verify achievements in their respective departments.

NFR-5.2: Students can only submit achievements for themselves.

NFR-5.3: Administrators can manage users within their assigned departments only.

NFR-5.4: Achievement verification must be completed within 7 days of submission.

NFR-5.5: Public visibility of achievements requires student consent.

6. Other Requirements

6.1 Database Requirements

DR-1: The database shall store user profiles, achievements, departments, and verification data.

DR-2: The database shall implement relational schema using PostgreSQL.

DR-3: The database shall maintain referential integrity between related entities.

DR-4: The database shall support full-text search capabilities.

DR-5: The database shall implement appropriate indexing for performance optimization.

6.2 System Administration Requirements

AR-1: The system shall provide tools for database management.

AR-2: The system shall include functionality for user role assignment.

AR-3: The system shall allow configuration of system parameters without code changes.

AR-4: The system shall provide monitoring tools for system health.

6.3 Backup and Recovery Requirements

BR-1: The system shall perform automated daily backups of all data.

BR-2: The system shall maintain backup history for at least 30 days.

BR-3: The system shall provide functionality for data restoration.

BR-4: The system shall implement recovery mechanisms in case of failure.